

Generation Interconnection System Impact Study Report

For Queue Project Q20-1611

Bell Shoals 138 kV

July 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1611. The Interconnection Customer proposes a 103.9 MW solar photovoltaic facility interconnecting at the Bell Shoals 138 kV substation in the Virginia area. The request was received on November 19, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 473 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,658,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
S Texa-Glen Oaks 138 KV	Brush Tavern-Rustburg 138 KV	123.5	173	12.4
Canby-Evington 138 KV line	Rustburg-Brush Tavern 138 KV	118.7	229	12.4
Glen Oaks-Weston 138 KV	Brush Tavern-Rustburg 138 KV	118.3	196	12.4
Arab-Dun Glenn 138 KV	Yaphank-S Colton Substation 138 KV	115.9	172	34.6
Canby-Weston 138 KV Line	Brush Tavern-Rustbug 138 KV line	115.9	215	12.4
Bell Shoals-Westlands 138 KV	S Colton-Yaphank 138 KV line	112.5	229	34.6
South Colton Substation-Yaphank Substation 138 KV	Bell Shoals-Westlands 138 KV Line	112.0	401	12.7
Dun Glenn-Milagro Substation 138 KV L	S Colton-Yaphnk 138 KV	110.7	197	34.6
Westlands-Milagro 138 KV	Yaphank-S Colton Substation 138 KV	109.9	215	34.6
Brush Tavern-Yaphank 138 KV	Bell Shoals-Westlands 138 KV	105.3	419	12.7
Pinas 9403-9405-Evington 138 KV	Brush Tavern-Rustburg 138 KV	104.0	268	12.4
South Colton-Pinas 9403-9405 138 KV line	Brush Tavern-Rustburg 138 KV	103.6	277	12.4
Craven County Wood Energy Lp Substation-Lone Tree 138 KV	Bear-Clover 345 KV	101.1	438	6.4
Craven County Wood Energy Lp-Lowmoor 138 KV	Bear-Clover 345 KV	101.1	433	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S Texa-Glen Oaks 138 KV Line	Rebuild line with higher-capacity conductor	\$2,206,000	10% (\$217,000)
Canby-Evington 138 KV line	Rebuild line with higher-capacity conductor	\$22,540,000	10% (\$2,220,000)
Weston-Glen Oaks 138 KV L	Rebuild line with higher-capacity conductor	\$3,781,000	10% (\$372,000)
Arab-Dun Glenn 138 KV	Rebuild line with higher-capacity conductor	\$3,055,000	45% (\$1,371,000)
Canby-Weston 138 KV Line	Rebuild line with higher-capacity conductor	\$17,142,000	10% (\$1,688,000)
Bell Shoals-Westlands 138 KV	Rebuild line with higher-capacity conductor	\$7,696,000	45% (\$3,453,000)
South Colton Substation-Yaphank Substation 138 KV	Rebuild line with higher-capacity conductor	\$6,950,000	8% (\$582,000)
Milagro-Dun Glenn 138 KV	Rebuild line with higher-capacity conductor	\$1,724,000	45% (\$773,000)
Westlands-Milagro 138 KV	Reconductor line	\$938,000	45% (\$421,000)
Brush Tavern-Yaphank 138 KV	Reconductor line	\$1,348,000	8% (\$113,000)
Pinas 9403-9405-Evington 138 KV	Reconductor line	\$1,075,000	10% (\$106,000)
South Colton-Pinas 9403-9405 138 KV line	Reconductor line	\$2,435,000	10% (\$240,000)
Craven County Wood Energy Lp Substation-Lone Tree 138 KV	Reconductor line	\$3,278,000	1% (\$44,000)
Craven County Wood Energy Lp-Lowmoor 138 KV	Reconductor line	\$4,272,000	1% (\$58,000)
	Total allocated to project		\$11,658,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.